

Solver-Supervised Graph Neural Surrogates for Parasitic Extraction and Layout Screening in PCB Planar Magnetics

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Abstract: PCB-embedded planar magnetics compress a converter but make winding parasitics unusually sensitive to foil placement, layer order, and lateral registration. Closed-form partial-element approximations are fast enough for a design loop, yet they can miss the same three-dimensional effects that the designer is trying to control. Full field extraction has the opposite problem. We train a geometry-aware graph surrogate for the inter-winding capacitance C_{ps} , the two free-space winding self-inductances L_p and L_s , and the winding mutual inductance M . A three-dimensional electrostatic finite-element solve supplies C_{ps} labels, and FastHenry supplies the inductive labels. On a held-out field-solver corpus, median errors are 2.63, 2.45, 3.31, and 1.75%, respectively. The replaced analytical labels differ from the same references by 51.5 to 266.6%. An independent Neumann-integral implementation places the learned inductive estimates within the spread between two solver formulations. A pairwise head addresses a separate design question: ordering variants of an unseen layout. It raises family-disjoint Spearman correlation from 0.27 for pointwise regression to 0.93 and, on lateral registration where the analytical capacitance is constant, reduces selection regret from 1176.1 to 3.5%.

We distinguish strict symmetry from predictive benefit. A corrected coordinate-update network is strictly $E(3)$ -equivariant on the encoded graph: maximum transformed-output and coordinate residuals are 3.92×10^{-7} and 1.41×10^{-7} . Its accuracy effect against a parameter-matched invariant distance-message network remains unresolved (+6.27%, 95% confidence interval $[-4.44, 17.68]\%$), and the interval fails a predeclared $\pm 2\%$ practical-equivalence criterion. Latency is measured at three boundaries. Median time is 0.33 ms/design for pre-collated batch throughput, 2.28 ms for a single forward pass, and 5.62 ms from raw layout through graph construction and inference, versus 3.66 s for the paired field solvers. The paired end-to-end speedup is $670\times$ (95% confidence interval 641 to $710\times$). The model is intended for solver-grade screening, not hardware-calibrated extraction; fabricated-board validation remains necessary.

Index Terms: Graph neural network, parasitic extraction, planar magnetics, partial-element equivalent circuit, finite element method, layout optimization, equivariance.

I. INTRODUCTION

Planar transformers and inductors replace wound wire with copper foils patterned across a printed-circuit stack. Their low profile and repeatable geometry suit high-density GaN and SiC converters, but the same construction couples the windings over broad overlapping areas. Inter-winding capacitance drives common-mode current, while self and mutual inductance shape leakage, ringing, and energy transfer. At high switching slew rates, moving one foil or changing one layer can alter the electromagnetic behavior enough to change the preferred layout [1–3].

The practical tension is familiar. Closed-form strip and parallel-plate models return an answer almost immediately. They are useful for hand calculation and early sizing, but their compactness comes from assumptions about current distribution, fringing, and overlap. Three-dimensional solvers resolve more of the geometry, at a cost that makes an exhaustive inner-loop search unattractive. A learned surrogate can occupy the gap only if its labels reflect the field solution. Matching a cheap approximation more accurately does not repair the

approximation.

We therefore tested the analytical labels before treating them as reference data. An analytical partial-element model generated the initial synthetic corpus. It produced a useful machine-learning benchmark, but comparison with field solvers exposed substantial label error. The revised pipeline regenerates capacitance with an electrostatic volume FEM and the inductance matrix with FastHenry. A message-passing graph neural network (GNN) is trained on those solver outputs. The graph representation preserves variable conductor count and exposes pair geometry to a shared message function. It also supports a sparse k -nearest-neighbor form when the all-pairs graph becomes unnecessarily expensive.

The experiments address five questions that arise in a design flow:

- How far are the convenient analytical labels from three-dimensional references, and does relabeling close that gap?
- Does graph structure earn its cost relative to pooled multilayer perceptrons and gradient-boosted trees?
- Can the model rank close variants within an unseen design family, not merely predict well over a heterogeneous test set?
- Is strict rigid-motion symmetry present in the implementation, and does the coordinate update improve accuracy or sample efficiency?
- What speedup remains when graph construction and batch size are stated explicitly?

Results depend on the task. Message passing helps with relational inductive targets and transfer across graph size, while pairwise training improves ordering. The strict equivariant coordinate update has no resolved advantage in the parameter-matched experiment, and its uncertainty interval is too wide to call the two models equivalent. At the small benchmark size, the dense graph is also slower than the analytical code. Its value is solver-level accuracy rather than speed relative to a lower-fidelity approximation.

Figure 1 summarizes the scope. The predicted vector is $[C_{ps}, L_p, L_s, M]$. Here M denotes winding mutual inductance in free space. It is not the ferrite-core magnetizing inductance, which depends on the magnetic circuit and is much larger in the operating component. Nor is the output a complete broadband RLGC matrix. These boundaries matter when interpreting both accuracy and deployment.

II. RELATED WORK

A. Parasitic Extraction for Planar Windings

The partial-element equivalent-circuit (PEEC) method converts electromagnetic interactions into circuit quantities [4]. Closed-form inductance expressions such as Grover’s formulas remain useful for rapid estimates [5], while FastHenry assembles and accelerates a three-dimensional partial-inductance solve [6]. For capacitance, FastCap introduced multipole-accelerated boundary-element extraction [7]; finite-element formulations offer an alternative when thin gaps or panel conditioning make a surface discretization difficult. Winding-resistance models trace back to Dowell and related treatments of skin and proximity effect [8, 9].

We do not propose a new field solver. The model approximates solver outputs over a constrained family of multilayer foil geometries.

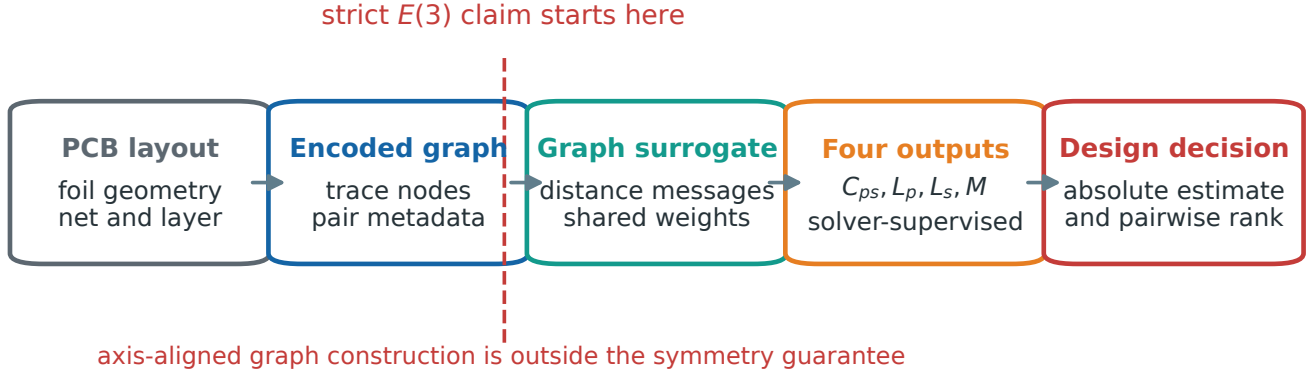


Fig. 1. From layout to design decision. The strict $E(3)$ statement applies after the layout has been encoded as a graph. The current graph builder uses axis-aligned overlap metadata and is not part of that guarantee. The regression and pairwise heads answer different questions: absolute parasitic estimation and ordering within a candidate family.

Numerical fidelity of the references and generalization of the surrogate are separate questions.

B. Graph Learning for Physical and EDA Problems

Message-passing networks represent a system as entities connected by learned interactions [10, 11]. They have been used to learn physical dynamics [12, 13], IC placement and sizing [14, 15], net-length estimation [16], and layout-dependent parasitics [17, 18]. Graphs are attractive here because conductor count varies and the targets accumulate pair interactions.

Rigid-motion-aware networks place a stronger constraint on those interactions. In an EGNN, scalar messages depend on invariant quantities such as squared distance, and coordinate updates are scalar-weighted relative vectors [19]. The construction is exact only if every learned message input respects the assumed group action. Raw Cartesian components passed to an unconstrained scalar MLP break that premise. We therefore test strict symmetry in the implementation rather than infer it from the architecture’s name.

C. Prediction Versus Ranking

An optimizer often needs the best candidate rather than a calibrated value for every candidate. Pointwise regression and learning to rank are not interchangeable objectives. RankNet and later ranking methods train on relative order or pairwise margins [20, 21]. The distinction is important when the between-family range is much larger than the variation among candidates in one family: high global R^2 can coexist with poor within-family ordering.

III. PROBLEM DEFINITION AND PHYSICAL SCOPE

A. Layout and Targets

A layout contains rectangular copper traces indexed by $i = 1, \dots, N$. Each trace has a net assignment, layer, center coordinate $x_i \in \mathbb{R}^3$, width w_i , length ℓ_i , thickness t_i , conductivity σ_i , and surrounding dielectric description. Primary and secondary turns are series-connected. We predict four extensive scalars,

$$y = [C_{ps}, L_p, L_s, M], \quad (1)$$

where C_{ps} is total primary-to-secondary capacitance, L_p and L_s are the free-space series self-inductances of the two windings, and M is their winding mutual inductance.

FastHenry returns a partial-inductance matrix rather than the series winding quantity directly. If $\mathcal{P}$ indexes primary traces, the consistent

composition is

$$L_p = \sum_{i \in \mathcal{P}} L_{ii} + 2 \sum_{\substack{i, j \in \mathcal{P} \\ i < j}} L_{ij}, \quad (2)$$

with analogous expressions for L_s and the cross-winding sum M . The surrogate is trained on these compositions, not on individual matrix entries.

The magnetic core is deliberately excluded. The operating transformer inductance includes a magnetizing term set by permeability, gap, and core geometry. A separate core calculation found that term to be roughly 37 to 73 times the free-space winding mutual over representative layouts. Thus M is a geometric winding quantity, not a substitute for magnetizing inductance.

B. Solver-Grade Rather Than Hardware-Grade

“Solver-grade” in this paper has a narrow meaning: held-out predictions are compared with the designated three-dimensional numerical references. It does not mean that a fabricated PCB was measured, that copper roughness and manufacturing tolerance were calibrated, or that the electrostatic model captures every layered-dielectric detail. A cross-solver inductance check reduces the risk of learning one program’s numerical idiosyncrasies, but two numerical formulations may still share modeling assumptions.

C. Symmetry Boundary

Let $Q \in O(3)$ and $t \in \mathbb{R}^3$. A rigid transform acts on encoded node coordinates as

$$x_i \mapsto Qx_i + t. \quad (3)$$

The four targets are invariant scalars. Updated coordinates inside an equivariant network should transform by the same action. Stored scalar metadata, including edge type and precomputed overlap, is held fixed in this test.

Strict $E(3)$ behavior therefore applies to the encoded graph. The layout-to-graph routine computes axis-aligned overlap before the transform. Rotating a raw layout and recomputing that overlap need not yield the same graph, so the guarantee does not extend to the entire raw-layout pipeline.

IV. REFERENCE SOLVERS AND DATA

A. Analytical Teacher and Its Audit

The initial teacher accumulates pairwise strip-inductance and capacitance approximations. Its speed made it useful for early architecture studies on 2000 layouts, and the GNN reproduced that teacher with

TABLE 1. Experimental corpora and the question each answers.

Experiment	valid	train	test	seeds	purpose
Field accuracy	332	265	67	1	four-target solver agreement
Strict $E(3)$	332	266	66	5	symmetry, accuracy effect, sample efficiency
Ranking	880	77 fam.	33 fam.	1	family-disjoint ordering
Frequency	136	109	27	1	$R_{ac}/R_{dc}(f)$ extension

$R^2 > 0.999$. The result measures teacher fidelity, not agreement with a field solution.

The discrepancy became visible when the same layouts were evaluated with independent references. Median analytical error was 51.5% for L_p , 53.6% for L_s , and 58.6% for M against FastHenry. For C_{ps} , an overlap formula based on dimensions but not lateral offset differed from the electrostatic FEM by 266.6%. The capacitance failure is especially instructive: two traces of equal dimensions can have identical analytical capacitance whether stacked or laterally disjoint, even though their true coupling changes sharply.

B. Inductance Reference

FastHenry discretizes each conductor cross-section and solves for the three-dimensional partial-inductance matrix [6]. The entries are composed according to (2). Every valid layout supplies L_p , L_s , and M . Positive mutual inductance and a completed matrix are required for inclusion.

An independent reference evaluates the Neumann double integral using a 5×2 filament subdivision. It shares neither the FastHenry executable nor its linear-system implementation. Across 70 held-out layouts, the GNN versus Neumann median errors are 6.61, 4.96, and 2.03% for L_p , L_s , and M ; FastHenry versus Neumann gives 4.95, 4.78, and 2.07%. The learned M therefore lies within the inter-solver discrepancy, while L_p retains a modest additional model error.

C. Capacitance Reference

The electrostatic reference solves

$$\nabla \cdot (\epsilon \nabla \phi) = 0 \quad (4)$$

on a tetrahedral volume mesh with primary and secondary conductors held at 1 and 0 V. The capacitance follows from stored field energy W ,

$$C_{ps} = \frac{2W}{V^2}. \quad (5)$$

The volume formulation was chosen after surface-panel experiments proved unstable at the approximately 0.1 mm inter-layer gaps. A stacked pair and a 60 mm-offset pair yielded 92.9 and 0.38 pF in the volume solve, whereas the position-blind analytical expression assigned both 44.3 pF.

D. Corpora and Splits

The field-accuracy experiment contains 332 valid designs and uses a 265/67 train/test split. The strict-equivariance ablation independently regenerates the same number of valid designs from 346 attempts and fixes a 266/66 split across every architecture and initialization. We report the experiments separately because their test indices differ. Training statistics are fitted only on the active training subset.

The layouts are synthetic eight-layer planar-LLC winding geometries. They vary trace count, dimensions, layer assignment, interleaving, and lateral registration. The field corpus reaches 70 trace nodes. No via, return-plane, connector, or measured process-variation model is included.

V. GRAPH SURROGATES

A. Encoded Graph

Each trace becomes a node. Invariant node scalars are width, length, thickness, log conductivity, relative permittivity, and normalized layer.

Directed edges connect either all trace pairs or the k nearest neighbors. For the audited invariant/strict comparison, learned messages receive overlap and coupling-type flags plus a distance recomputed from coordinates. Raw Δx , Δy , and Δz components are not sent to an unconstrained scalar MLP.

The dense graph has $O(N^2)$ edges and mirrors the all-pairs reference. The sparse graph has $O(Nk)$ edges. Since magnetic and electric coupling decay with separation, local neighborhoods retain much of the predictive signal. The asymptotic saving matters only after graph construction and inference dominate other fixed costs.

B. Invariant Distance-Message Network

Let $h_i^{(\ell)}$ denote the hidden scalar state at layer ℓ , a_{ij} invariant edge metadata, and $r_{ij}^{(\ell)} = \|x_i^{(\ell)} - x_j^{(\ell)}\|^2$. A message and node update are

$$m_{ij}^{(\ell)} = \phi_m(h_i^{(\ell)}, h_j^{(\ell)}, r_{ij}^{(\ell)}, a_{ij}), \quad (6)$$

$$h_i^{(\ell+1)} = \text{LN}\left[h_i^{(\ell)} + \phi_h\left(h_i^{(\ell)}, \text{mean}_j m_{ji}^{(\ell)}\right)\right]. \quad (7)$$

The invariant baseline keeps $x_i^{(\ell+1)} = x_i^{(\ell)}$. Four layers are followed by mean, max, and signed-log-sum pooling. Their concatenation enters a four-output head. The sum channel is useful because the targets are extensive.

C. Strict Coordinate-Update Network

The strict model uses the identical scalar path and adds

$$x_i^{(\ell+1)} = x_i^{(\ell)} + \text{mean}_j \left[\frac{x_i^{(\ell)} - x_j^{(\ell)}}{\sqrt{r_{ij}^{(\ell)}} + 1} c(m_{ij}^{(\ell)}) \right], \quad (8)$$

where $c(\cdot)$ is an invariant learned scalar bounded by a scaled hyperbolic tangent. Because the multiplier is invariant and the displacement is a relative vector, (8) is equivariant to rotations, reflections, and translations. Scalar pooling makes the graph output invariant.

The strict width-96 network has 310,763 trainable parameters. The primary invariant comparison uses width 102 and 307,945 parameters, a 0.91% difference. A width-96 invariant network with 273,127 parameters supplies a sensitivity comparison. Widening only the baseline is not a perfect control for optimization, so both comparisons are retained.

D. Pairwise Ranking Head

For candidates a and b from one design family, the ranking head produces scores s_a, s_b and minimizes a margin loss,

$$\mathcal{L}_{\text{rank}} = \max(0, \gamma - z_{ab}(s_a - s_b)), \quad (9)$$

where $z_{ab} \in \{-1, 1\}$ encodes the solver ordering. Families, rather than individual layouts, are assigned to train or test, which prevents variants of the same base geometry from leaking across the split.

VI. EXPERIMENTAL PROTOCOL

A. Training and Metrics

All networks use AdamW, learning rate 2×10^{-3} with cosine decay, weight decay 10^{-5} , batch size 32, gradient-norm clipping at 5, and SmoothL1 loss on standardized signed-log($1 + |y|$) targets. Scalar feature and target moments are estimated from training data only. Coordinates are divided by one isotropic training-derived scale; separate axis scales would destroy exact rotational symmetry.

The primary EGNN metric is a per-design macro error,

$$e_i = \frac{1}{4} \sum_{q=1}^4 |\log(1 + \hat{y}_{iq}) - \log(1 + y_{iq})|. \quad (10)$$

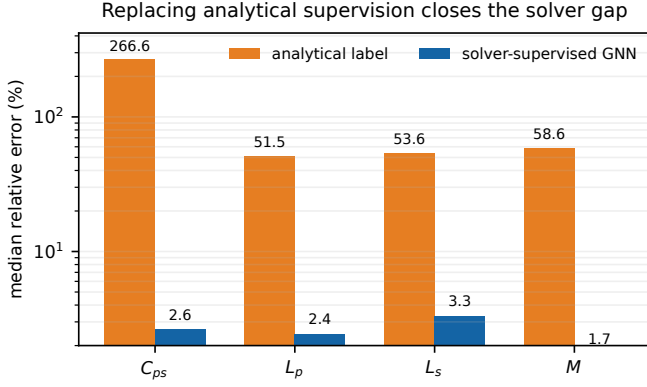


Fig. 2. Median relative error against the designated three-dimensional reference. Analytical values are teacher errors; GNN values are held-out prediction errors after relabeling. The logarithmic axis keeps the four improvements visible without hiding their different scales.

It avoids giving the largest-magnitude target automatic control of the comparison. Per-target R^2 and median relative error remain secondary descriptive metrics.

The strict ablation fixes one test set and five initialization/training-order seeds. Nested training sizes are 25, 50, 100, 200, and the full 266 designs. Uncertainty uses 10,000 paired hierarchical bootstrap resamples: seeds are resampled first and held-out designs second. The relative effect is strict minus invariant; negative values favor strict EGNN. Practical equivalence was defined before inspecting results as a 90% interval entirely inside $[-2, 2]\%$. A directional practical effect requires the 95% interval to clear the corresponding $\pm 2\%$ boundary.

B. Symmetry Test

Each full-data strict model is evaluated under proper rotations, reflections, and translations. The test compares invariant outputs and equivariant coordinates against their transformed expectations. Relative tolerance is 2×10^{-5} . The five models contribute 200 transformed cases. Failure of any full-data seed blocks a strict-symmetry claim.

C. Latency Boundaries

Latency can change by nearly an order of magnitude without changing the network, simply by moving the timer. We therefore report:

1. *pre-collated batch throughput*: one prepared test batch divided by batch size;
2. *single forward*: a prepared graph evaluated alone;
3. *raw-layout end to end*: graph construction, normalization, collation, forward pass, and inverse target transform;
4. *paired solver*: FastHenry plus the electrostatic FEM for the same design.

The primary speedup pairs each design’s solver time with its end-to-end GNN time and bootstraps the median ratio. Throughput is useful for screening a large candidate bank but is not substituted for batch-one latency.

VII. RESULTS

A. Field-Solver Accuracy

The choice of labels has a larger effect than the architecture changes studied here. Figure 2 compares analytical discrepancy with held-out solver-supervised GNN error. Error falls for all four targets. Capacitance decreases from 266.6 to 2.63%; median error for the three inductive quantities decreases from a range of 51.5 to 58.6% to a range of 1.75 to 3.31%. Table 2 gives the corresponding R^2 values.

The capacitance R^2 is lower than the inductive values even though

TABLE 2. Held-out field-solver accuracy (332 valid layouts; 265/67 split).

Target	reference	median error	R^2
C_{ps}	electrostatic FEM	2.63%	0.9622
L_p	FastHenry	2.45%	0.9939
L_s	FastHenry	3.31%	0.9953
M	FastHenry	1.75%	0.9954

its typical relative error is similar. Squared residuals and the high-capacitance tail have a strong effect on R^2 , whereas the median relative error describes a typical design. The two metrics therefore expose different failure modes.

B. Why Use a Graph?

Table 3 compares the GNN with four tree ensembles built from the same pooled 27-dimensional summary on an identical 332-layout corpus with a 67-layout test set. The GNN reaches 2.63% median C_{ps} error and $R^2 = 0.962$, versus 2.9% and 0.942 for LightGBM. A graph is not necessary to obtain a competitive scalar capacitance predictor.

Relational quantities separate the models. Tree R^2 remains between 0.878 and 0.920 on the inductances, while the GNN reaches 0.994 to 0.995. On a graph-size extrapolation split, tree $R^2(M)$ falls to between 0.112 and 0.324; the GNN retains 0.943. The shared message function applies without redesign when trace count changes. A pooled MLP likewise reproduced scalar capacitance but degraded toward $R^2 \approx 0.88$ on L_p and L_s . Together, the baselines attribute the inductive improvement to pairwise geometry rather than parameter count alone.

C. Strict $E(3)$: Verified Symmetry, Unresolved Accuracy Effect

Each of the five strict models passes the transformation gate. Across 200 cases, the maximum relative residual is 3.92×10^{-7} for graph outputs and 1.41×10^{-7} for updated coordinates, more than an order of magnitude below the 2×10^{-5} tolerance. The measured residuals verify strict $E(3)$ behavior on the encoded graph, including reflections.

Symmetry verification does not imply a predictive gain. On the full training set, the primary error is 0.028911 for strict EGNN and 0.027205 for the parameter-matched invariant model. The relative change is $+6.27\%$, with 95% CI $[-4.44, 17.68]\%$. The 90% interval $[-2.81, 15.94]\%$ is not contained in the predeclared $\pm 2\%$ equivalence region. Neither improvement nor practical equivalence is established.

The same-width sensitivity arm is also inconclusive: strict EGNN error is 6.98% higher, 95% CI $[-1.08, 14.98]\%$. Because this baseline has fewer parameters, the result should not replace the parameter-matched primary analysis. It does show that the coordinate update does not produce a robust advantage merely by adding capacity. Figure 3 displays both comparisons.

Sample efficiency is similarly unresolved. The normalized area under error versus $\log_2(n_{\text{train}})$ is 0.07% lower for strict EGNN, 95% CI $[-1.94, 1.81]\%$. Individual training sizes change direction, which is why a learning curve and paired seeds are more informative than a selected single run.

D. Ranking and Design Regret

The pointwise field-accuracy model orders variants of unseen families at only $\rho = 0.27$. A dedicated pairwise head raises median family-disjoint ρ to 0.929 across 33 test families. The gap is not a paradox: global error is dominated by differences among base designs, while an optimizer compares much smaller changes within one family.

Layer order and lateral registration produce different ranking behavior. For layer-order interleaving, the analytical model already sees vertical spacing and reaches $\rho = 0.78$; the GNN reaches 0.81, and both select low-regret candidates (5.4 and 5.8%). For lateral registration, the analytical overlap expression is constant in every test family and cannot rank at all. The GNN reaches $\rho = 0.95$. When asked to select the minimum- C_{ps} candidate, analytical screening incurs 1176.1% regret

TABLE 3. Tree baselines and GNN on the identical field-solver split. Parentheses contain median relative error in percent. Rank ρ is reported only where a family-disjoint ranking experiment exists; size transfer is $R^2(M)$.

Model	C_{ps}	L_p	L_s	M	rank ρ	size transfer
Random Forest	.462 (12.9)	.887 (24.8)	.914 (15.9)	.880 (9.3)	n/a	.112
XGBoost	.927 (3.8)	.878 (16.3)	.908 (16.5)	.914 (7.8)	n/a	.231
LightGBM	.942 (2.9)	.893 (23.9)	.890 (20.2)	.918 (7.1)	.088	.247
CatBoost	.892 (4.7)	.896 (24.5)	.893 (20.4)	.920 (8.1)	n/a	.324
GNN	.962 (2.63)	.994 (2.45)	.995 (3.31)	.995 (1.75)	.929	.943

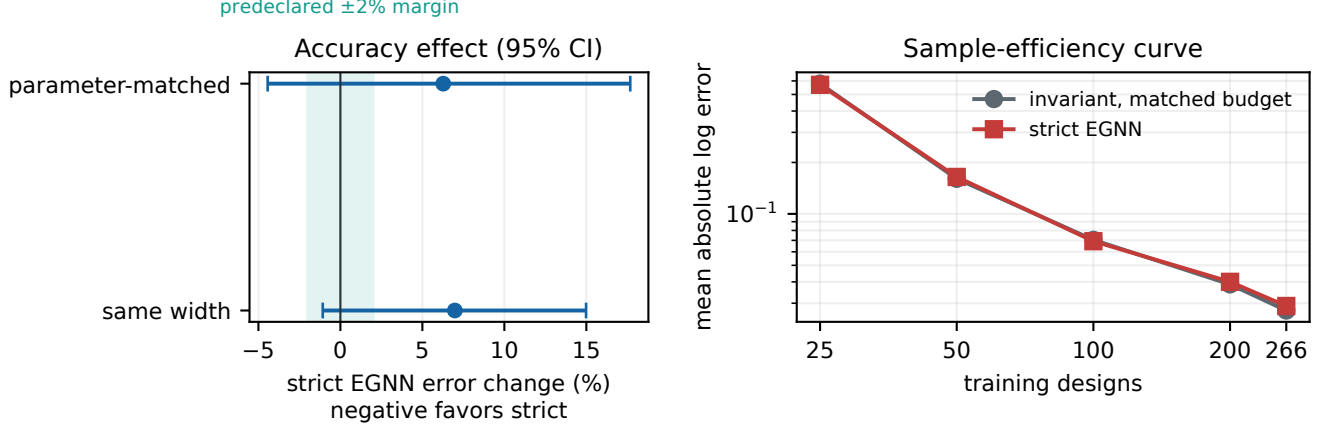


Fig. 3. Strict-coordinate-update ablation. Left: relative primary-error effects with paired hierarchical-bootstrap 95% intervals; the shaded band is the predeclared practical-equivalence margin. Right: five-seed mean learning curves on one fixed split. Exact encoded-graph symmetry is verified, but its accuracy and sample-efficiency effects are unresolved in the parameter-matched comparison.

TABLE 4. Controlled strict- $E(3)$ ablation (five seeds, fixed 266/66 split). Positive error change favors the invariant model.

Comparison	effect	95% CI
Full, parameter matched	+6.27%	$[-4.44, 17.68]$
Full, same width	+6.98%	$[-1.08, 14.98]$
Sample-efficiency AULC	-0.07%	$[-1.94, 1.81]$
Output residual, max		3.92×10^{-7}
Coordinate residual, max		1.41×10^{-7}

TABLE 5. Latency and speedup definitions. All values are medians per design unless noted.

Timed boundary	value
GNN, pre-collated batch	0.331 ms
GNN, single prepared graph	2.285 ms
GNN, raw-layout end to end	5.617 ms
Paired field solvers	3.659 s
Paired end-to-end speedup	$670.2 \times$
95% speedup interval	$[640.8, 710.4] \times$

relative to the FEM optimum, similar to random selection at 1126.2%; GNN screening reduces it to 3.5% (Fig. 4). Three-dimensional supervision changes the selected design for this second layout variable.

E. Latency and Speedup

The latency depends on where timing begins (Fig. 5). Pre-collated batch throughput is 0.33 ms/design, but a prepared graph evaluated alone takes 2.28 ms and the raw-layout path takes 5.62 ms. The paired FastHenry and FEM reference has a 3.66 s median, 3.75 s mean, and 4.66 s 95th percentile. The phrase “about 5 s” summarizes the upper part of this distribution rather than an exact constant.

The primary comparison uses the per-design ratio between paired solver time and raw-layout end-to-end inference. Its median is $670.2 \times$, with 95% bootstrap interval $[640.8, 710.4] \times$. For an interactive batch-one path, this is the relevant boundary. A historical 1.17 ms figure came from pre-collated batch throughput; the same protocol produced 0.675 ms on the current node and runtime stack, confirming that the absolute number depends on hardware and software. Combining the rounded 5 s solver estimate with the historical throughput gives approximately $4300 \times$, but that calculation does not include the raw-layout end-to-end boundary.

Against the analytical implementation, the dense graph is slower at the benchmark’s mean size. Sparse k -NN inference changes the asymptotic edge count from $O(N^2)$ to $O(Nk)$ and crosses the un-accelerated all-pairs analytical implementation near $N = 80$ in the measured scaling study. The crossover concerns runtime; below that

size, the reason to use the surrogate is solver-grade accuracy.

VIII. FREQUENCY-RESOLVED EXTENSION

The lumped vector does not describe frequency-dependent copper loss. As a separate task, the primary $R_{ac}/R_{dc}(f)$ curve is sampled at 17 logarithmic frequencies from 10 kHz to 100 MHz. FastHenry with cross-section subdivision supplies the reference. A 17-output GNN trained on 109 of 136 boards achieves 0.37% median curve error and $R^2 = 0.995$ on 27 held-out boards.

Figure 6 shows the mean curve rising from approximately one to six across the band. The predicted and reference curves nearly overlap. The graph representation can therefore support a frequency-indexed output, but this experiment uses a smaller corpus and its own split; its metric is not merged with the four-target field-accuracy result.

IX. DISCUSSION

A. What the Graph Contributes

The evidence does not support a blanket statement that a graph is superior for every output. A well-tuned tree ensemble is competitive for typical C_{ps} error because the capacitance target can be strongly correlated with pooled size and overlap statistics. The graph earns its complexity on the inductive quantities, size transfer, and pairwise decision task. Those are precisely the cases in which relations among traces cannot be compressed safely into fixed global moments.

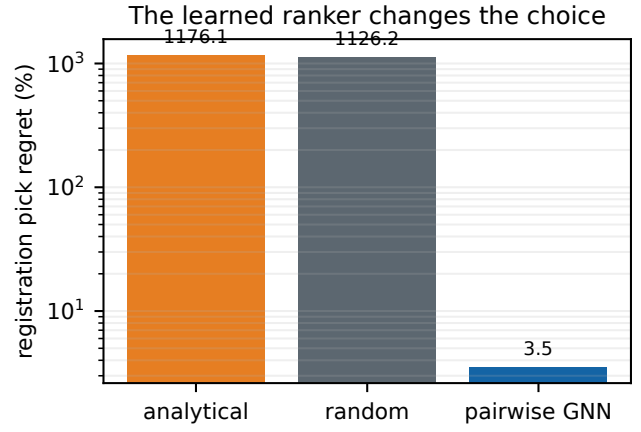
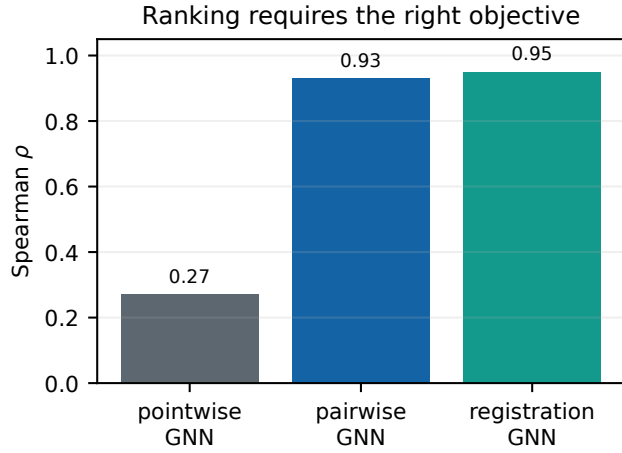


Fig. 4. Ranking behavior. A pairwise objective resolves fine within-family order that pointwise regression misses. On lateral registration, the analytical capacitance is position-blind; its selected layout is no better than random, whereas the learned ranker nearly recovers the FEM optimum.

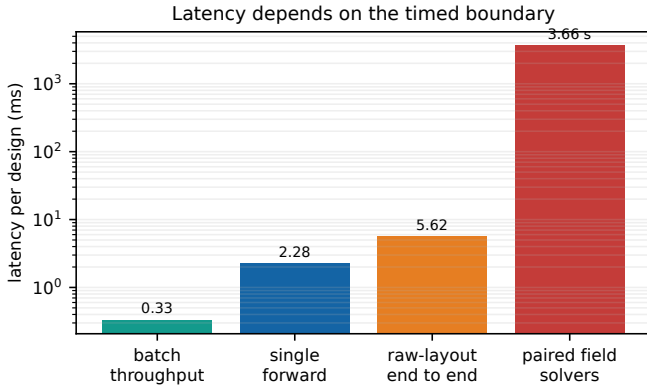


Fig. 5. Measured latency at four timer boundaries. The vertical axis is logarithmic. Pre-collated throughput is useful for a candidate bank, while raw-layout end-to-end time is the conservative batch-one deployment measure.

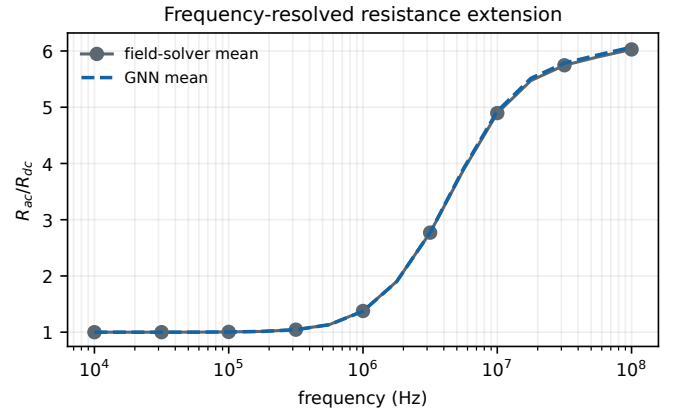


Fig. 6. Mean held-out $R_{ac}/R_{dc}(f)$ curve for the separate frequency-resolved task. The 17-output model attains 0.37% median curve error over 27 test boards.

Exact symmetry is an implementation guarantee, not evidence of lower prediction error. It removes coordinate-frame dependence from the encoded-graph model and defines a transformation contract. The present field corpus resolves neither an accuracy benefit over a strong invariant distance-message network nor practical equivalence. More seeds or a benchmark with arbitrary three-dimensional orientations would narrow the observed uncertainty.

B. Deployment in a Design Loop

A practical workflow uses the GNN to screen a broad candidate bank, the pairwise head to order close variants, and a field solve to confirm a small finalist set. The surrogate is not the final sign-off tool. Batched throughput estimates the cost of a candidate bank, whereas 5.62 ms measures an isolated raw-layout query.

The implementation uses PyTorch without PyG or DGL. The dependency choice simplifies embedding and licensing, although it is not a scientific accuracy claim. Sparse graphs should be selected when trace count is high enough for graph construction and all-pairs messages to matter. For the modest boards in the accuracy corpus, dense connectivity remains a reasonable reference architecture.

C. Limitations

The study has no fabricated-board anchor. FastHenry, the electrostatic FEM, and the independent Neumann calculation establish numerical consistency within a modeled geometry. They do not include connector fixtures, solder, copper roughness, or manufacturing distributions. Impedance or resonant measurements on a controlled PCB coupon are

needed before claiming hardware accuracy.

The geometry family is intentionally narrow: rectangular foils in an eight-layer planar winding, without vias, return planes, complex cutouts, or arbitrary curved routing. The strict model is verified only on the graph produced from this family. Extending the raw-layout encoder while preserving a well-defined symmetry action is a separate problem.

The reported inductances are winding quantities in free space. The model does not learn the ferrite magnetic circuit or its nonlinear, frequency-dependent permeability. A separate FEM sanity check reproduces an EER40 GP95 datasheet A_L within roughly 4 to 6% at representative production gaps, but that experiment validates the core solver rather than this GNN.

The capacitance FEM uses an effective dielectric description. Layered anisotropic dielectrics, frequency-dependent loss tangent, and full-wave coupling are outside its scope. These omissions matter most when the surrogate is moved from layout ranking to absolute EMI prediction.

X. REPRODUCIBILITY AND ARTIFACT ORGANIZATION

The repository contains the dataset generators, graph and model code, solver interfaces, experiment outputs, and the complete LaTeX source for this manuscript. The numerical source for every new figure is a machine-readable JSON file beside the plotting script. Figure generation is deterministic. A clean build invokes the figure script and runs the LaTeX toolchain from the manuscript directory.

Heavy reference generation is intentionally separated from docu-

ment building. Recreating the 332-design field corpus requires repeated three-dimensional solves and should run on a managed compute system. Rebuilding the figures and PDF consumes only released result files. External FastHenry and capacitance-solver binaries are not redistributed; their interfaces and build instructions are provided.

XI. CONCLUSION

Replacing analytical labels with solver outputs reduces held-out median error to a range of 1.75 to 3.31% across C_{ps} , L_p , L_s , and M . An independent Neumann formulation bounds the remaining inductive discrepancy. Graph structure contributes most on relational targets, size transfer, and fine layout decisions. A pairwise objective improves within-family ordering and nearly recovers the optimal lateral-registration choice where the analytical model is blind.

The corrected EGNN supplies strict $E(3)$ behavior on encoded graphs, but its predictive effect remains unresolved against a parameter-matched invariant model. Latency depends on the stated boundary: the raw-layout path is 5.62 ms/design and gives a paired end-to-end median speedup of $670\times$ over the evaluated field workflow. The result supports solver-grade screening on synthetic geometries. Measured boards, richer layout primitives, layered dielectric references, and joint core-winding models remain outside the present validation.

A. FEATURE AND SYMMETRY CONTRACT

Table 6 records the representation used in the controlled invariant/strict comparison. An earlier exploratory model passed all seven edge channels, including Cartesian vector components, to an unconstrained MLP. It provides a counterexample: an EGNN-style coordinate update does not rescue scalar invariance when its message already depends on a coordinate-frame-specific input.

The coordinate normalization uses one isotropic scale. Standardizing x , y , and z with separate factors maps a rotation to a non-orthogonal transformation in normalized space. Data augmentation has the same constraint: rotating coordinates while leaving a raw relative-vector edge channel unchanged tests a different graph rather than the claimed group action.

The symmetry gate uses relative residual

$$\delta_y = \frac{\|f(QX + t) - f(X)\|_2}{\|f(X)\|_2 + \varepsilon} \quad (11)$$

for graph output and

$$\delta_x = \frac{\|X_L(QX + t) - (QX_L(X) + t)\|_F}{\|QX_L(X) + t\|_F + \varepsilon} \quad (12)$$

for coordinates after the last layer. The largest observed values are reported in Table 4; no rounded prediction percentage is used as the strictness criterion.

B. SEED-LEVEL ARCHITECTURE RESULTS

Table 7 exposes the five full-data primary errors rather than presenting only a pooled effect. Strict EGNN is better for one seed and worse for four. The variation explains the broad hierarchical interval and prevents a single favorable initialization from being presented as an architectural result.

At 50 training designs, strict EGNN is practically worse by 11.07% with 95% interval [3.27, 19.74]%. At 25, 100, and 200 designs the intervals cross zero. These pointwise learning-curve analyses are secondary to the preselected AULC summary, but they show that sample efficiency is not monotonically improved by coordinate updates.

C. TEACHER FIDELITY AND SOLVER ACCURACY

The 2000-layout analytical-label corpus remains useful for controlled architecture debugging. On its held-out set, dense message passing

reaches $R^2 = 0.9998, 0.9996, 0.9995$, and 0.9997 for C_{ps} , L_p , L_s , and M . A two-layer model is similar to the four-layer model, while removal of relative geometry roughly doubles inductance RMSE. A sparse k -NN model retains $R^2 \geq 0.992$ in the principal four-target comparison.

Those values should be read as fidelity to a deterministic analytical teacher. They answer whether the network can learn the extensive pairwise calculation, not whether the calculation is physically accurate. The field-corpus results in Table 2 answer the second question. Keeping the two tables separate avoids the misleading inference that $R^2 > 0.999$ applies against the three-dimensional solvers.

The failed capacitance-BEM route is also retained in the release. Uniform-panel estimates varied by factors of six to forty as panel density changed at the thinnest dielectric gaps, and the resulting corpus produced negative held-out R^2 . Adaptive refinement improved conditioning but increased solve time substantially. The volume FEM was selected because its conductor regions and dielectric volume yielded stable geometry sensitivity over the required gap range. Its selection reflects the numerical behavior in this geometry family, not a universal preference over BEM.

D. ADDITIONAL TIMING INTERPRETATION

The paired solver median is 3.659 s, while its mean is 3.750 s and 95th percentile 4.656 s. The spread comes mainly from the electrostatic mesh and solve; FastHenry is the smaller component for these layouts. Reporting a distribution matters for two reasons. First, a projected bank runtime based on a rounded 5 s constant is conservative near the median but not a direct measurement of every candidate. Second, ratios of independent medians and medians of paired ratios need not agree.

The primary $670.2\times$ result is therefore the median of paired per-design solver/end-to-end ratios. Its 95% interval $[640.8, 710.4]\times$ re-samples designs. The pre-collated throughput is reported separately and applies only when graph preparation is outside the timed loop or amortized over a candidate bank. Hardware, thread count, warm-up, batch size, repeat count, and timer boundary remain necessary context when comparing latency with another system.

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TABLE 6. Feature contract for the controlled architecture comparison. “Invariant” refers to the encoded-graph $E(3)$ action defined in Section III-C.

Component	channels	transformation	use in both arms
Node scalars	width, length, thickness, $\log \sigma$, ϵ_r , normalized layer	invariant	encoded to hidden scalar state
Node coordinates	normalized x, y, z with one isotropic scale	vector	used to recompute squared distance
Edge metadata	overlap, capacitive flag, inductive flag	invariant by test definition	encoded once and held fixed under graph transform
Distance	$\ x_i - x_j\ ^2$	invariant	recomputed inside every message layer
Coordinate update	bounded scalar weight times $x_i - x_j$	equivariant	strict arm only
Readout	mean, max, signed-log-sum	invariant	shared graph-level pooling

TABLE 7. Full-data primary error by initialization seed. Lower is better.

Seed	invariant, matched	strict EGNN
42	0.030449	0.032618
43	0.027932	0.030436
44	0.029287	0.026443
45	0.023085	0.027717
46	0.025271	0.027341
Mean	0.027205	0.028911

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